EE 451: Communications Systems

Final Exam — Practical Skills Rubric (W3USR Amateur Radio Station)

Exam date: Thursday, May 21, 2026, 12:45–2:45 PM **Practical worth:** 35 of 100 total exam points (35%) **Where:** W3USR Amateur Radio Station **Format:** One student at a time. Dr. Frissell administers and grades while Dr. Shahrouzi proctors the written portion. Plan on roughly 10–15 minutes per student.

What you are being evaluated on

The skills below are exactly what we have been practicing in the W3USR sessions during L24–L27. The exam is hands-on: you sit at the IC-7610 with the instructor, perform the tasks, and explain what you are doing as you go. There are no surprise topics — every item maps to something we have done in class.

Bring nothing. Reference materials at the station (great-circle map, ITU phonetic chart, DXCC prefix list) are available to all students.

Logistics

- The IC-7610 will be configured in a known state at the start of each student's session
- Default band is **20 m USB**; 40 m LSB is the fallback if propagation is poor
- The instructor is the licensed control operator for all transmissions
- If band conditions prevent finding a real station to work, the instructor may simulate the other station for items 2 and 8
- A logbook page will be available for item 7

Rubric (35 points total)

#	Skill	Max pts
1	Station configuration	3
2	Tune in an SSB station and copy the callsign	5
3	Construct a callsign for a chosen country	3
4	Read the S-meter and give an RST	4
5	Use the antenna rotor with a great-circle bearing	4
6	ITU band identification	3
7	Q-signal interpretation (3 codes)	3
8	Complete a basic QSO (call or answer CQ)	6
9	Connect one task to course theory	4
Total		**35**

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1. Station configuration (3 pts)

Starting from a reset state, set the IC-7610 to receive on 20 m phone:

- Mode = **USB** (1 pt)
- Frequency anywhere in the **14.150–14.350 MHz** phone segment (1 pt)
- Correct antenna selected, AF gain audible, RF gain reasonable (1 pt)

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2. Properly tune in an SSB station and copy the callsign (5 pts)

Tune the IC-7610 across the phone segment, find an active SSB station, and adjust the dial so the audio sounds natural and intelligible — no “Donald Duck” pitch shift, voice on the right sideband, fine-tuned for clarity. Once the signal is clean, write down the next callsign you hear in standard letters and digits (e.g., `W3USR`) — the same way you would log it in a real QSO. You may listen as many times as needed for the callsign; the emphasis here is on *tuning skill*. The phonetic-alphabet skill is tested separately when *you* speak your own callsign in item 8.

Tuning (3 pts):

- 3 pts — signal is properly tuned: voice sounds natural, intelligible, no pitch distortion; sideband and fine-tuning correct
- 2 pts — signal is mostly tuned but slightly off — small pitch correction still needed
- 1 pt — signal is recognizable but noticeably off-frequency or distorted; substantial adjustment still needed
- 0 pts — signal not properly tuned (Donald-Duck pitch, wrong sideband, no usable signal selected)

Callsign copy (2 pts):

- 2 pts — fully correct callsign (any number of listens)
- 1 pt — partial copy (majority of letters/digits correct)
- 0 pts — incorrect or no answer

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3. Construct a callsign for a country of your choice (3 pts)

Using the **DXCC list provided at the station**, pick any country you like, construct a plausible callsign for that country, and announce it in proper ITU phonetics. The structure is: a valid country prefix from the DXCC list, followed by a digit and a 1–3 letter suffix of your invention.

- 1 pt — Country paired with a valid prefix from the DXCC list (e.g., **DL** for Germany, **JA** for Japan, **VK** for Australia, **G** for England, **LU** for Argentina)
- 1 pt — Callsign has plausible structure: valid prefix + digit + 1–3 letter suffix (e.g., DL5ABC, JA3XY, VK7QQ, G4MNO)
- 1 pt — Callsign is spoken correctly using the full ITU phonetic alphabet (e.g., “Delta Lima Five Alfa Bravo Charlie”)

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4. Read the S-meter and give an RST (4 pts)

For a station you can hear, give an RST report and briefly explain it.

- 2 pts — readable report (R) consistent with how well you can copy the audio
- 2 pts — strength (S) consistent with the IC-7610 S-meter
- Partial credit for one digit correct, the other not
- Bonus check (no extra pts): be ready to explain what each digit measures if asked

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5. Use the antenna rotor with a great-circle bearing (4 pts)

The instructor names a country (e.g., Japan, Argentina, the UK). You:

1. Determine the great-circle bearing from W3USR using the wall map or online tool (2 pts)
2. Rotate the antenna to that bearing using the rotor controller (2 pts)

- Bearings within ±20° of correct earn full credit on step 1 (a typical Yagi half-power beamwidth is ~50–70°)

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6. ITU band identification (3 pts)

Given the operating frequency you are listening on (e.g., 14.2 MHz, or 7.2 MHz on the 40 m fallback):

- Name the ITU band designation (HF / VHF / UHF / etc.) (1 pt)
- State the wavelength range that defines that band (1 pt)
- State one propagation characteristic of that band (1 pt)

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7. Q-signal interpretation (3 pts)

The instructor uses three Q-signals in short example sentences (e.g., *"My QTH is Scranton"*, *"Lots of QRN tonight"*, *"Let's QSY to 14.250"*). For each, state in plain English what the Q-signal means (1 pt each).

Q-signals you should know: **QTH, QRM, QRN, QSB, QRZ, QSO, QSY, QSL, QRT, QRP, QRO**.

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8. Complete a Basic QSO by Calling or Answering CQ (6 pts)

Complete a basic QSO using either approach:

- **Answering CQ:** Tune the band, find a station calling CQ, and reply.
- **Calling CQ:** Find a clear frequency, call CQ properly ("CQ CQ CQ this is Whiskey Three Uniform Sierra Romeo …"), and work whoever answers.

The instructor is the control operator and will give you a go-ahead before you key the microphone. Either path counts equally.

- Initiate the contact correctly (1 pt) — proper CQ call OR proper reply to someone else's CQ, with your callsign in phonetics
- Copy the other station's callsign correctly (1 pt)
- Give a signal report and your QTH (1 pt)
- Receive and acknowledge the other station's information (1 pt)
- Sign off correctly with "73" and the W3USR callsign (1 pt)
- Log the contact: date, UTC time, frequency, mode, their callsign, RSTs sent and received (1 pt)

If propagation prevents a real contact, the instructor will play the role of the other station and the same rubric applies. Logging is graded either way.

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9. Connect one task to course theory (4 pts)

Pick **one** of the operations you performed and explain the underlying communications-systems concept from earlier in the semester, **with at least one quantitative or technical detail** (a formula, a number, a dB value, a bandwidth, etc.). Then be ready for **one short follow-up question** from the instructor.

A separate **Theory Connection Study Sheet** lays out the connections you should be ready to draw — phonetics □ channel coding, RST □ SNR in dB, skip propagation □ frequency-selective channel, antenna gain □ link budget, USB □ BFO recovery of SSB, panadapter □ FFT of I/Q, FT8 □ 8-FSK, Costas array □ thumbtack autocorrelation, ITU bands □ propagation regimes, Q-signals □ source coding, QSB □ time-varying multipath channel. Pick one ahead of time and rehearse it. **You are not limited to the study-sheet examples.**

Scoring:

- 4 pts — Clear, correct connection between an operating task and a course concept; includes a correct quantitative or technical detail; handles the follow-up question competently
- 3 pts — Correct connection with a minor error in the technical detail OR a weaker follow-up answer
- 2 pts — Connection is broadly correct but vague (no quantitative detail) and/or follow-up is incomplete
- 1 pt — Connection is on the right track but technically wrong in important ways
- 0 pts — No meaningful connection or fundamentally incorrect

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How to study for this practical

- Practice the ITU phonetic alphabet (table below) until you can spell anything without thinking
- For item 3: pick a country you'd like to use ahead of time and rehearse a callsign for it in phonetics; the DXCC list at the station gives you the valid prefixes (sample blocks below)
- Memorize the eleven Q-signals (table below)
- Know the ITU band designations (table below) and the rough propagation character of each
- Know the RST scale (table below) — what each digit means and where on the S-meter "S9" sits
- Run through the CQ exchange in your head: who calls, who replies, what each station says

- For item 9: read the **Theory Connection Study Sheet**, pick one connection that resonates, and rehearse it out loud — including the quantitative detail and a likely follow-up question

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Quick Reference

The tables below cover everything you need to study for the practical. The same materials (phonetic chart, DXCC prefix list, great-circle map) are also posted at the station for use during the exam.

ITU Phonetic Alphabet

Letter	Word	Letter	Word	Letter	Word
A	Alfa	J	Juliett	S	Sierra
B	Bravo	K	Kilo	T	Tango
C	Charlie	L	Lima	U	Uniform
D	Delta	M	Mike	V	Victor
E	Echo	N	November	W	Whiskey
F	Foxtrot	O	Oscar	X	X-ray
G	Golf	P	Papa	Y	Yankee
H	Hotel	Q	Quebec	Z	Zulu
I	India	R	Romeo		

Digits: spoken as their English names — “zero, one, two, three, four, five, six, seven, eight, niner.” Use *niner* for nine to disambiguate from German *nein*.

Q-Signals (the eleven you must know)

Code	Meaning
QTH	“My location is …” / “What is your location?”
QRM	Man-made interference (other signals, electrical noise)
QRN	Natural noise (lightning, atmospherics)
QSB	Signal fading
QRZ	“Who is calling me?”
QSO	A two-way contact
QSY	“Change frequency to …”
QSL	“I confirm receipt” / a confirmation card
QRT	“I am ceasing transmission” / shutting down
QRP	Low power
QRO	High power

ITU Band Designations

Band	Frequency	Wavelength	Propagation character
MF	300 kHz – 3 MHz	1 km – 100 m	Ground wave by day, sky wave at night
HF	**3 – 30 MHz**	**100 m – 10 m**	**Ionospheric skip — long distance**
VHF	30 – 300 MHz	10 m – 1 m	Line-of-sight, sporadic E
UHF	300 MHz – 3 GHz	1 m – 10 cm	Strict line-of-sight, satellites
SHF	3 – 30 GHz	10 cm – 1 cm	Line-of-sight, weather attenuation

20 m phone (14.150–14.350 MHz) lives in HF. The 40 m fallback (7.150–7.300 MHz) is also HF.

RST Scale

R — Readability (1–5):

Value	Meaning
5	Perfectly readable
4	Readable with practically no difficulty
3	Readable with considerable difficulty
2	Barely readable, occasional words distinguishable
1	Unreadable

S — Strength (1–9):

Value	Meaning
9	Extremely strong (calibration reference: **S9 = –73 dBm**, each S-unit ≈ 6 dB)
7	Moderately strong
5	Fairly good signals
3	Weak signals
1	Faint, barely perceptible

A “5-by-9” report means perfectly readable at S9 strength. Reports above S9 are given as “S9 + 10,” “S9 + 20,” etc. (each being 10 or 20 dB above S9).

(The T digit applies only to CW. Voice contacts on SSB give just R and S.)

Sample DXCC Prefix Blocks

The station’s full DXCC list is what you’ll use during item 3. These are common blocks for orientation:

Country / Region	Prefix(es)
United States	K, N, W, AA–AL
Canada	VE, VA, VO, VY
United Kingdom	G, M, 2
Germany	DL, DA–DR
France	F
Japan	JA–JS
Australia	VK
Brazil	PY, PT
Argentina	LU
South Africa	ZS
Israel	4X, 4Z
Switzerland	HB9

Grading sheet (instructor copy)

#	Skill	Max	Score	Notes
1	Station configuration	3		
2	Tune in an SSB station and copy the callsign	5		
3	Construct callsign + speak in phonetics	3		
4	RST report	4		
5	Rotor + great-circle bearing	4		
6	ITU band identification	3		
7	Q-signals (3)	3		
8	QSO (call/answer CQ) + log	6		
9	Theory connection	4		
	Total	**35**		

Student name: ____________________________ Date: _____________

Instructor signature: ____________________________